Individual Course Project

SmartPET

Pet Shop Management System

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COMP 8090SEF Data Structures And Algorithms

OOP-Based Application Development

2026

Overview

What is SmartPET?

A Python application for managing a pet store, demonstrating OOP concepts through practical implementation.

Core Functions

✓ Managing Pet Inventory
✓ Tracking Customers
✓ Processing Sales

Three Core Functions

SmartPET –Pet Shop Management System

Managing Pet Inventory

Add, view, and manage pets with details including name, age, price, breed, type etc.

Tracking Customers

Maintain customer records with contact information and purchase history

Tracking Customers

Sell pets to customers with inventory updates and revenue tracking

System Design

Pet Customer → **PetShop Controller**

Design: Data models (Pet, Customer) are managed by the PetShop controller class using composition

OOP Concepts used in SmartPET

Concept	Application
Class and Object	Three classes: Pet, Customer, PetShop
Encapsulation	Private attributes with getters / setters
Class Attributes	Shared ID counter and pet counter
Static Methods	Price formatting
Magic Methods	__str__, __eq__, __len__
Modular Programming	Four files by functionality

Application Demo

```
============================================================
         SmartPET - Pet Shop Management System
============================================================
  1. View Shop Information
  2. Add Pet to Inventory
  3. View All Pets
  4. View Available Pets
  5. Add Customer
  6. View All Customers
  7. Process Sale
  8. View Sales History
  9. Generate Sample Data
  0. Exit
```

- ✓ Manage pet inventory
- ✓ Track customers
- ✓ Process sales

Project Structure

main.py
Entry point & User Interface

models.py
Pet and Customer classes

petshop.py
PetShop controller class

utils.py
Helper functions

Thank you

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